

IT 360 Final Project Report

Prepared by

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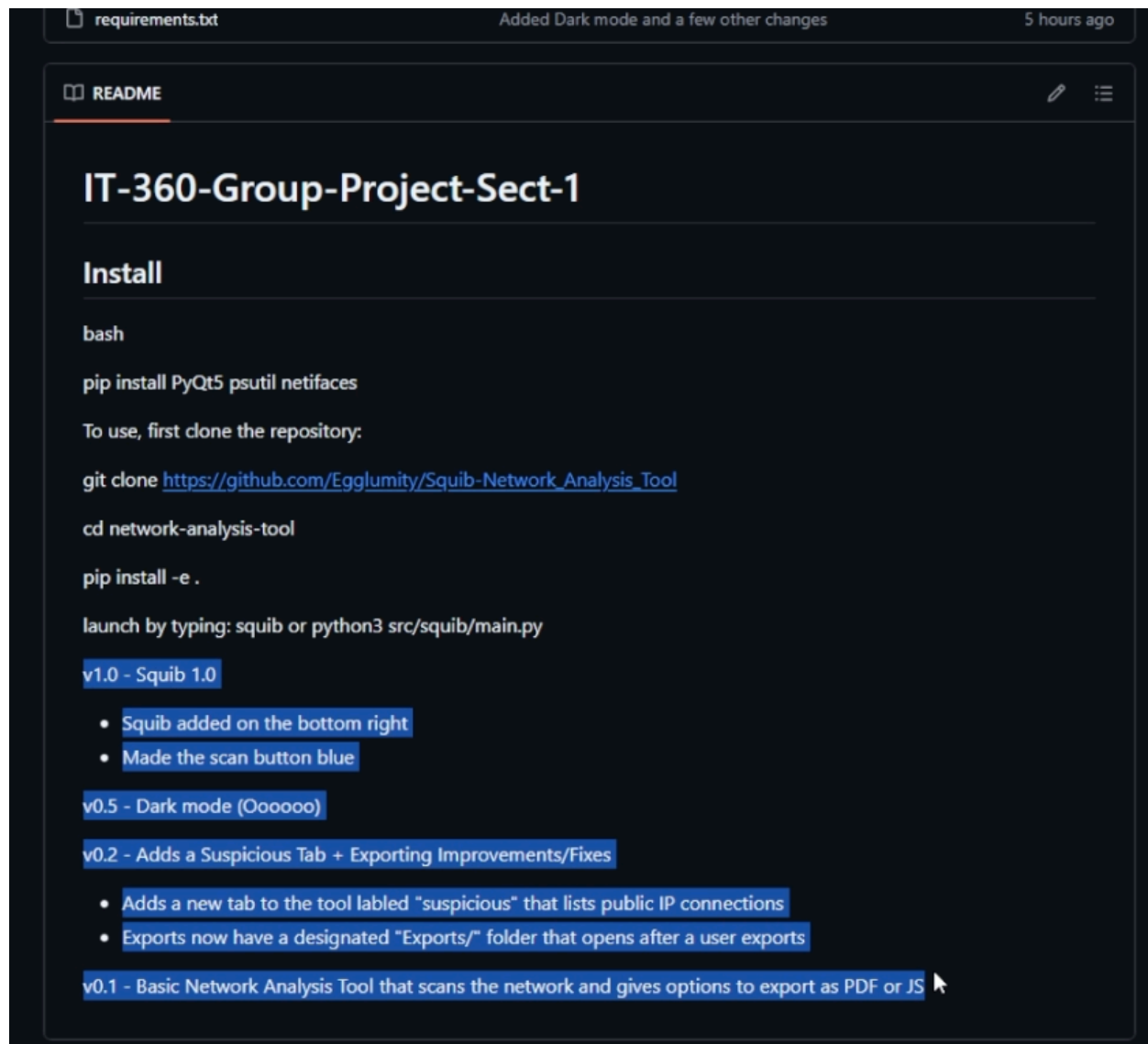
1. Introduction

Squib is a network analysis tool that we developed for the purpose of making it easier to scan for open ports or risky connections. This project allowed us to understand more about network vulnerabilities and how to use python to run automation on our system. We also learned a lot about github and how repositories are used and worked on. While we encountered many obstacles while working to make this project meet our expectations, we were able to learn and figure out our way through it with the help of research and AI tools. Our main goal for Squib is for it to be able to run network scans on a host pc and return its results in a clean, easy to read way. With countless hours of research and late nights, we were able to make a tool we were happy with.

If we were to implement more updates in the future, some of these examples include returning possible exploits that your pc is at risk of, as well as integrating an AI into our tool to return better results.

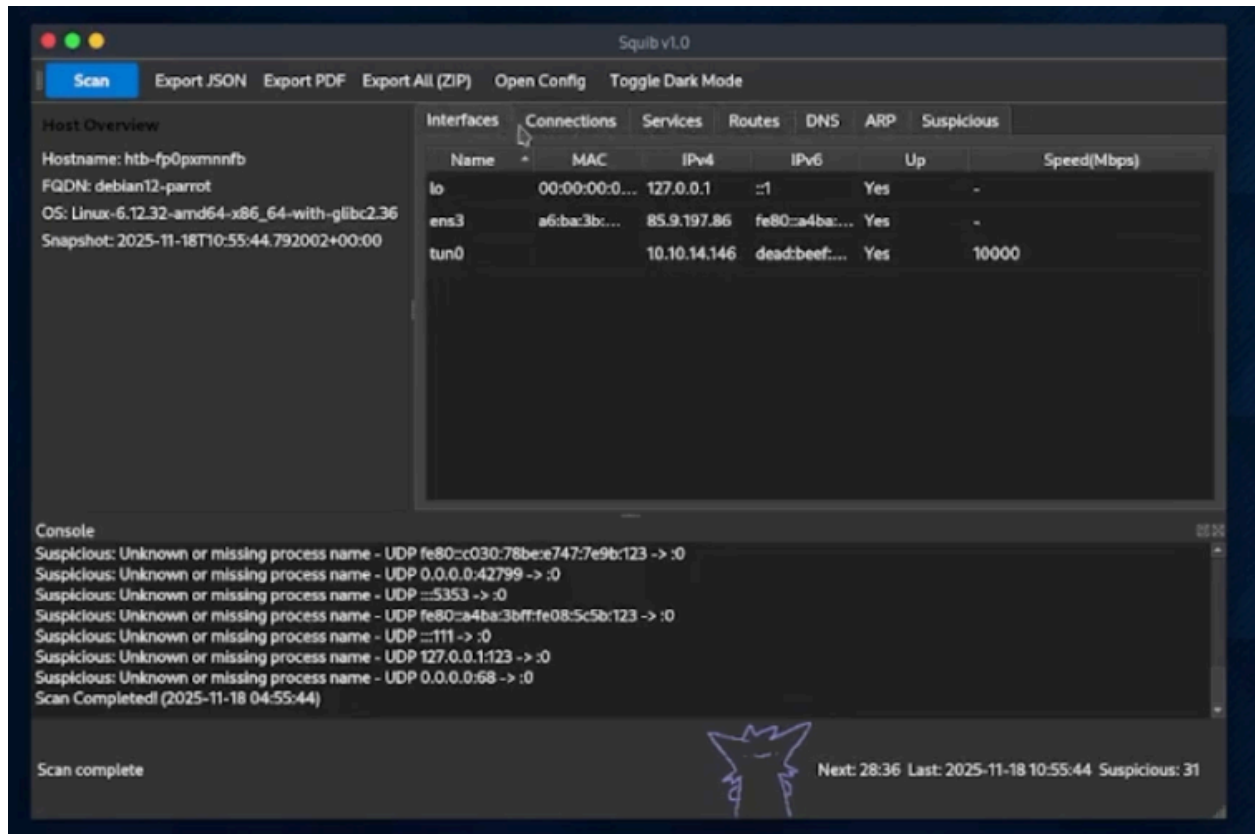
2. Technical Implications

When making our tool for the first time, some problems we ran into was getting the tool to launch correctly, some libraries might've not been installed correctly, making us take time in setting all of that up, but now our README file says exactly what needs to be installed prior to using the tool.

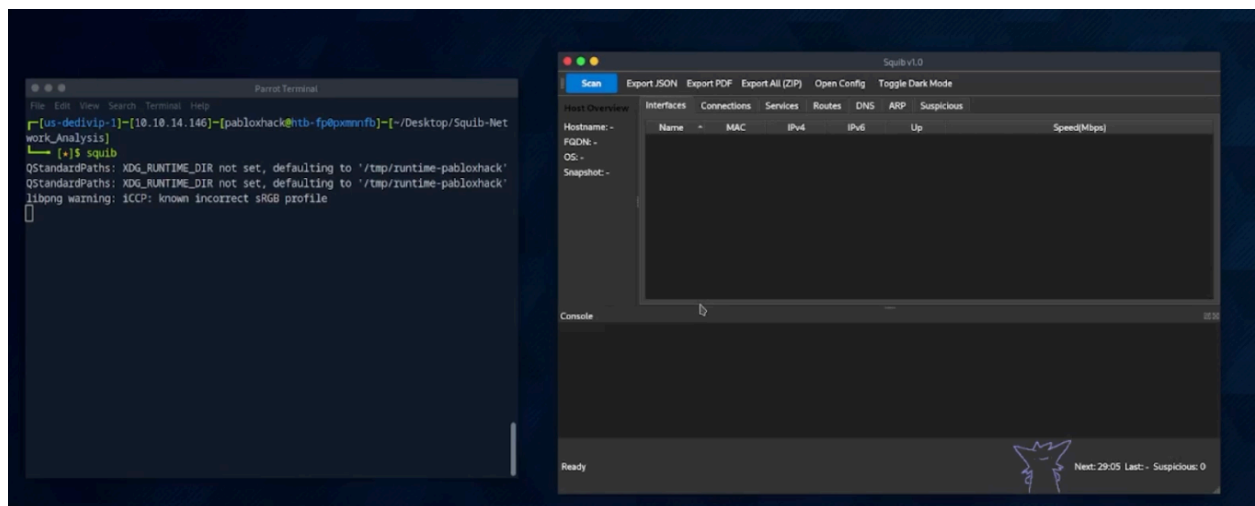


3. Results

Our original objective for our tool was to have something that could scan a network and spot suspicious connections such as public connections or open ports. Originally it was only supposed to scan when the user clicked scan, but we implemented a config file that lets the user decide how often it wants the tool to run a scan, letting it run fully automatically in the background. We also have it export the results to a folder every time, so everything is kept.



After we finished with everything, we wanted our tool to be recognizable, so we came up with its name (Squib), and made a mascot for it that appears on the GUI. We thought this was way better than just naming our tool "Network Analysis Tool".



4. Conclusion

If we planned on integrating AI into our tool later on, we'd have to find a way to manage the token usage, and which tool we would even use. We could have the user use an AI of their choice, but that would make for a more complex setup while our tool is meant to be an easy to setup and use program.

To secure export logs to prevent manipulation or public viewing, we could add hash verification to provide authenticity and integrity. For the JSON files we could encrypt it with AES and add a password prompt on every access request.

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Squib's Suspicious tab could be more in depth, such as scanning the versions of common programs and seeing if they're updated and open to vulnerability.

We learned a lot during this project, especially how to use Github to make sharing and collaborating on our project simpler. We were able to use Python and its GUI library to create a nice looking GUI for our project. We also learned a lot about Linux and its ports and connections, what should be considered a suspicious connection, and how to export the findings to a folder.